

Factorial experiments and Multifactor ANOVA

Chapter 5

Example: battery life

Let us assume we are interested in the **life** (Response variable) of a particular kind of Battery

After a team discussion, you decide to consider **two factors**:

A Electrode Material (**Qualitative-Nominal Factor**)

B Operating Temperature °F (**Quantitative-Interval Factor**)

Pay attention: ANOVA will always consider a Quantitative Factor as a Nominal one

Questions:

1. Have material and temperature any **effect** on the the battery life?
2. Could we select some **optimal couple** of levels?
3. Could we choose an optimal material **unregarding the working temperature** in order to maximize the battery life (**robust** product)?

Design of the Experiment

We must decide:

1. Number of factor levels
2. Number of replicates
3. Design typology 
4. How to conduct the experiment

How do we select the Design typology?

- **“Best-guess” Experiments**
- **OFAT “One Factor at A Time”**
- **Factorial Experiment** and its variations by Fisher

R. A. Fisher and his legacy



- In **Oct 1919**, Fisher joined **Rothamsted Experimental Station**.
- His assignment was to “**examine our data and elicit further information that we had missed.**” (by John Russell, Station Director)
- By **1926** (a mere 7 years 😊), Fisher invented **AN**alysis **Of VA**riance and **D**esign **Of E**xperiments as new methods to design and analyze **agricultural experiments**.

Fisher's Principles in DoE

- **Factorial scheme**
- **Replication:** to assess process variation
- **Randomization/Blocking.** (to counteract the effect of unknown noising factors).

“**Block what you can and Randomize what you cannot.**”
- Originally motivated by **agricultural experiments** (quite slow experiments), Thereafter they have been widely used for any physical experiments.

Factorial Experiments

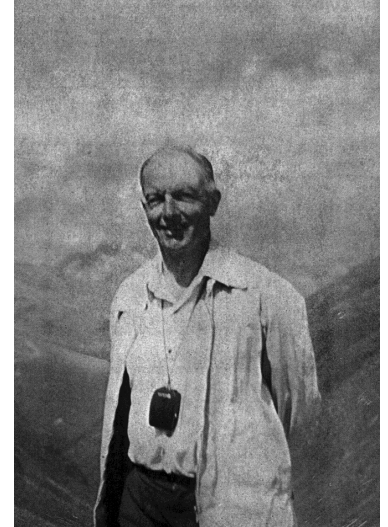
- Factorial arrangement to accommodate factorial structure of treatment/block, by **Fisher** (1926) . Originally called “**complex experiments**”.
- Major work on **factorial design** by **F. Yates** (1935, 1937), and **fractional factorials** by **D. Finney** (1945); both worked with Fisher.
- Major development after WWII for applications to industrial experiments, by the Wisconsin School, **G.E.P. Box** and co-workers (J. S. Hunter, W. G. Hunter).



1902-1994
Frank Yates



1919-2013
G.E.P. Box



1917-2018
David J. Finney

Guiding Principles for Factorial Effects

- **Effect Hierarchy Principle:**
 - Lower order effects are more important than higher order effects;
 - Effects of same order are equally important (before experimenting).
- **Effect Sparsity Principle:**
 - The number of relatively important effects is small.
- **Effect Heredity Principle:**
 - for an interaction to be significant, at least one of its parent factors should be significant

Wu-Hamada “Experiments: Planning, Analysis, and Optimization”,
John Wiley & Sons

Two-Factor Factorial Design

Table 5-2 General Arrangement for a Two-Factor Factorial Design

		Factor <i>B</i>			
		1	2	...	<i>b</i>
Factor <i>A</i>	1	$y_{111}, y_{112},$ $\dots, y_{11n}$	$y_{121}, y_{122},$ $\dots, y_{12n}$		$y_{1b1}, y_{1b2},$ $\dots, y_{1bn}$
	2	$y_{211}, y_{212},$ $\dots, y_{21n}$	$y_{221}, y_{222},$ $\dots, y_{22n}$		$y_{2b1}, y_{2b2},$ $\dots, y_{2bn}$
	$\vdots$				
	<i>a</i>	$y_{a11}, y_{a12},$ $\dots, y_{a1n}$	$y_{a21}, y_{a22},$ $\dots, y_{a2n}$		$y_{ab1}, y_{ab2},$ $\dots, y_{abn}$

a levels of Factor *A*;

b levels of Factor *B*;

n replicates (usually constant across the factor levels)

full balanced CRD - Completely Randomized Design

We have **ab treatments**

Total number of runs **N=nab**

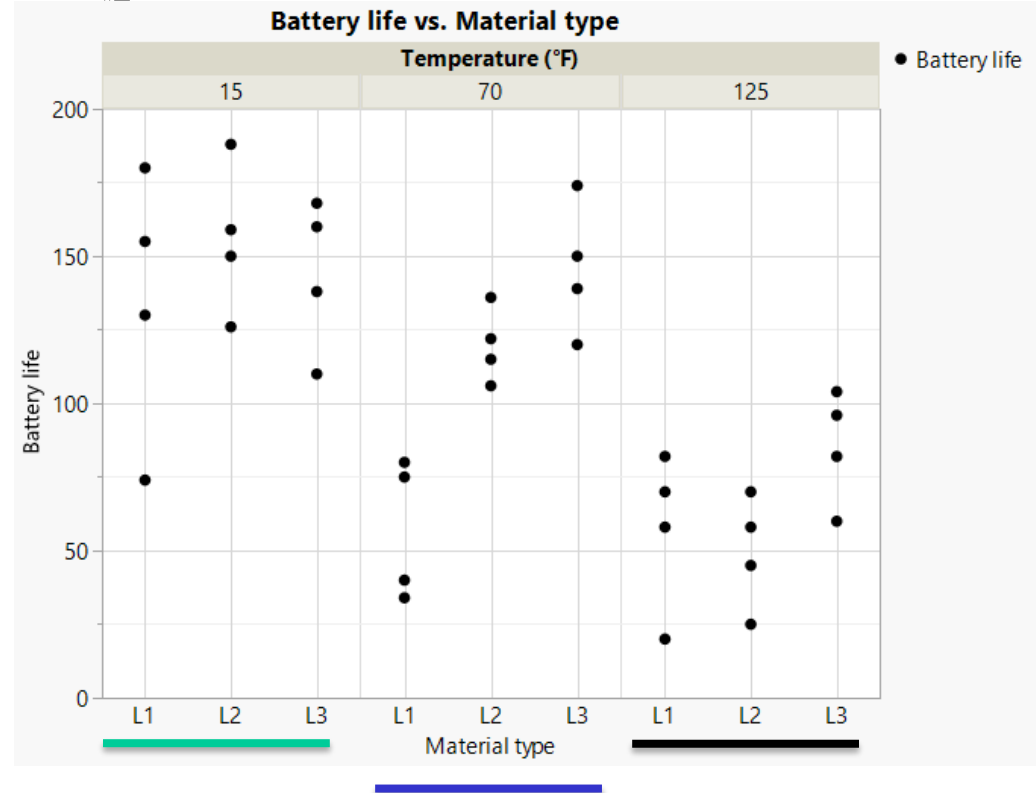
EDA (Battery life)

Table 5-1 Life (in hours) Data for the Battery Design Example

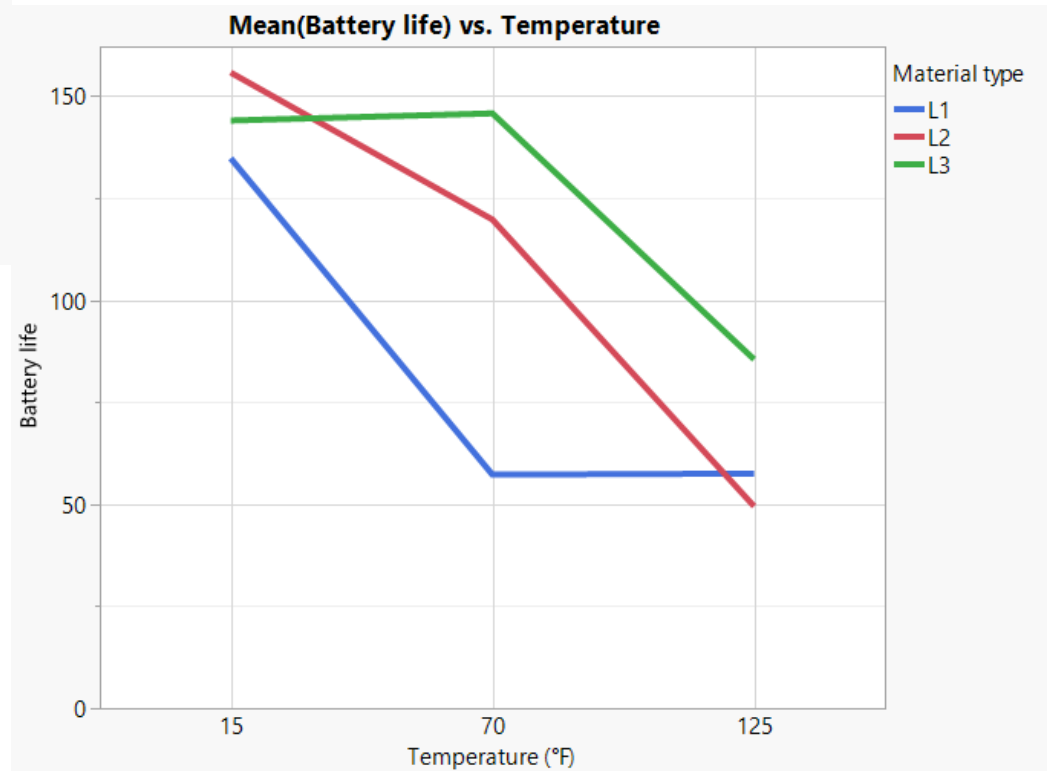
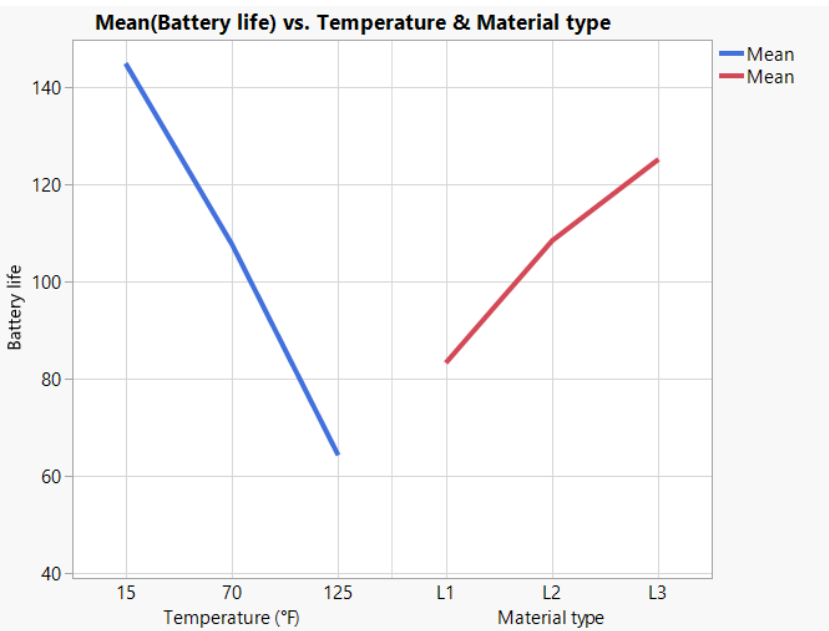
Material Type	Temperature (°F)					
	15		70		125	
1	130	155	34	40	20	70
	74	180	80	75	82	58
2	150	188	136	122	25	70
	159	126	106	115	58	45
3	138	110	174	120	96	104
	168	160	150	139	82	60

9 treatments

Discuss the plot



Main Effect and Interaction Plots



We have several possible statistical models:

I. Fixed Effects

II. Random Effects

III. Mixed

Model I (Fixed Effects, Complete Model)

$$y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$$

**Deterministic
component**

**Random
component**

$$\mu_{ij} = \mu + \tau_i + \beta_j + \tau\beta_{ij}$$

$\mu \equiv$ General Mean

$\tau_i \equiv$ Main Effect of Factor A

$\beta_j \equiv$ Main Effect of Factor B

$\tau\beta_{ij} \equiv$ Interaction Effect

$$\varepsilon_{ijk} \sim NID(0, \sigma^2)$$

Remark: We can use other models (mean, regression)

Reduced Models

$$y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon$$

$$y_{ijk} = \mu + \tau_i + \varepsilon = \mu_i + \varepsilon$$

$$y_{ijk} = \mu + \beta_j + \varepsilon = \mu_j + \varepsilon$$

$$y_{ijk} = \mu + \varepsilon$$

Remark: from now on we will not use the indices for the **random component** of the model

Estimate of the model parameters (OLS)

$$\text{Min} \sum_{i,j,k} \varepsilon_{ijk}^2 \rightarrow \text{Min} \sum_{i,j,k} \left[y_{ijk} - \left(\mu + \tau_i + \beta_j + \tau\beta_{ij} \right) \right]^2$$

$$1 \rightarrow \mu : y_{...} = abn\hat{\mu} + bn \sum_i \hat{\tau}_i + an \sum_j \hat{\beta}_j + n \sum_{i,j} \hat{\tau}\hat{\beta}_{ij}$$

$$a \rightarrow \tau_i : y_{i..} = bn\hat{\mu} + bn\hat{\tau}_i + n \sum_j \hat{\beta}_j + n \sum \tau\hat{\beta}_{ij}$$

$$b \rightarrow \beta_j : y_{.j.} = an\hat{\mu} + n \sum_i \hat{\tau}_i + an\hat{\beta}_j + n \sum_i \hat{\tau}\hat{\beta}_{ij}$$

$$ab \rightarrow \tau\beta_{ij} : y_{ij.} = n\hat{\mu} + n\hat{\tau}_i + n\hat{\beta}_j + n\hat{\tau}\hat{\beta}_{ij}$$

Plus **Additional constraints** to solve the Normal Equations

$$\sum_i \hat{\tau}_i = 0 \quad \sum_j \hat{\beta}_j = 0 \quad \sum_i \hat{\tau}\hat{\beta}_{ij} = 0 \quad \sum_j \hat{\tau}\hat{\beta}_{ij} = 0$$

Normal Equation Solutions

$$\hat{\mu} = \frac{y_{...}}{abn} = \bar{y}_{...}$$

$$\hat{\tau}_i = \bar{y}_{i..} - \hat{\mu} = \bar{y}_{i..} - \bar{y}_{...}$$

$$\hat{\beta}_j = \bar{y}_{.j.} - \hat{\mu} = \bar{y}_{.j.} - \bar{y}_{...}$$

$$\hat{\tau\beta}_{ij} = \bar{y}_{ij.} - \hat{\tau}_i - \hat{\beta}_j - \hat{\mu} = \bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}$$

Estimate (FIT or PREDICTED) of y_{ij}

If the interaction is **significant**

$$\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \hat{\tau\beta}_{ij} = \bar{y}_{ij}.$$

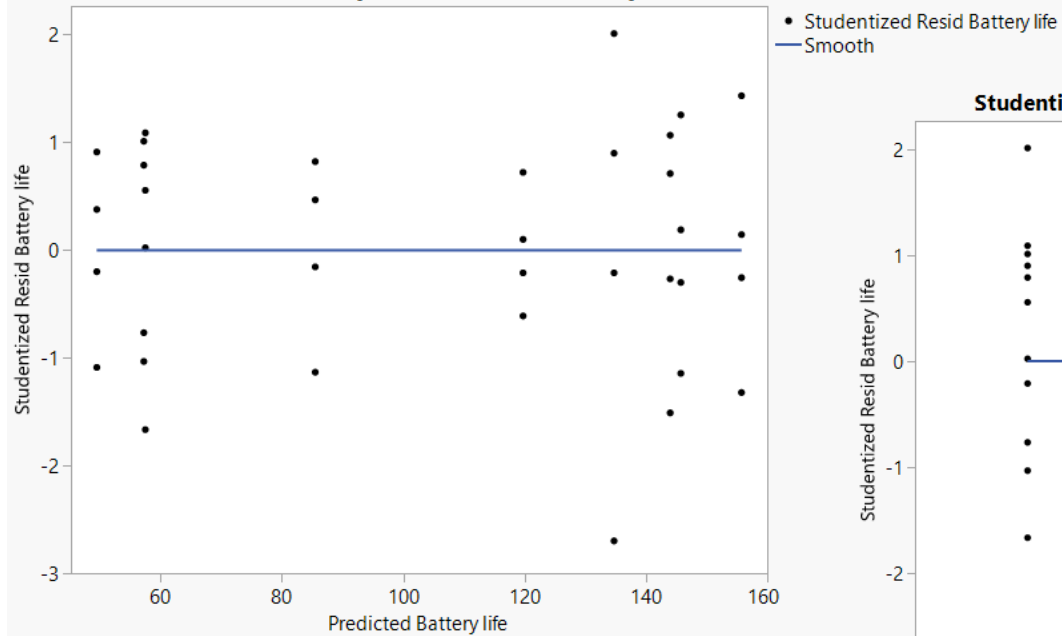
If the interaction is **not significant**

$$\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j = \bar{y}_{i..} - \frac{\bar{y}_{...}}{2} + \bar{y}_{.j.} - \frac{\bar{y}_{...}}{2}$$

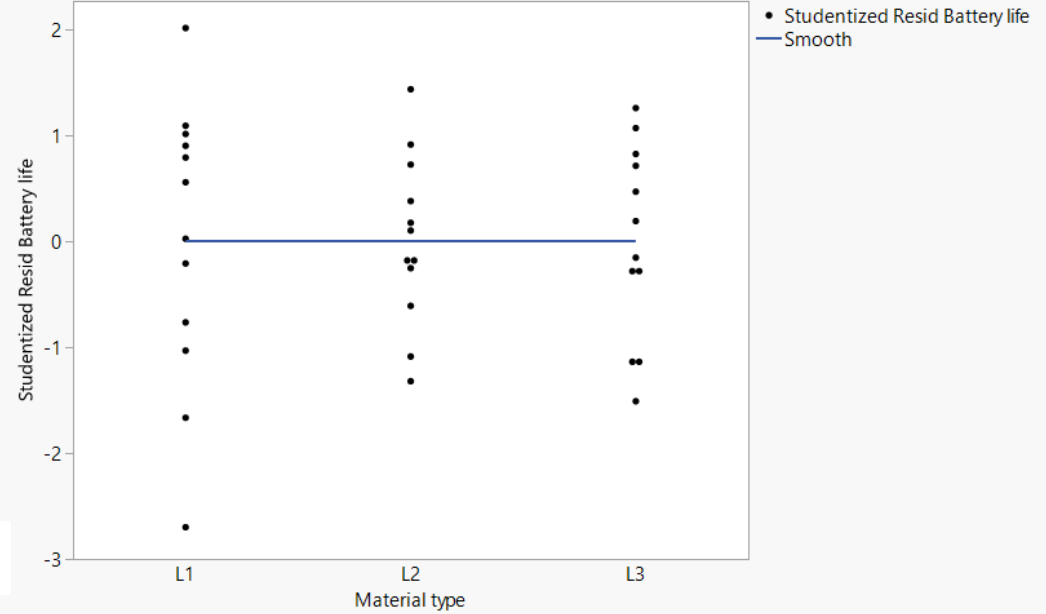
Usually, the right model is **not** important for testing hypotheses about residuals!

$$e_{ijk} = y_{ijk} - \hat{y}_{ij} \quad \longrightarrow \quad e_{ijk}^* = \frac{e_{ijk}}{\sqrt{\hat{V}(e_{ijk})}}$$

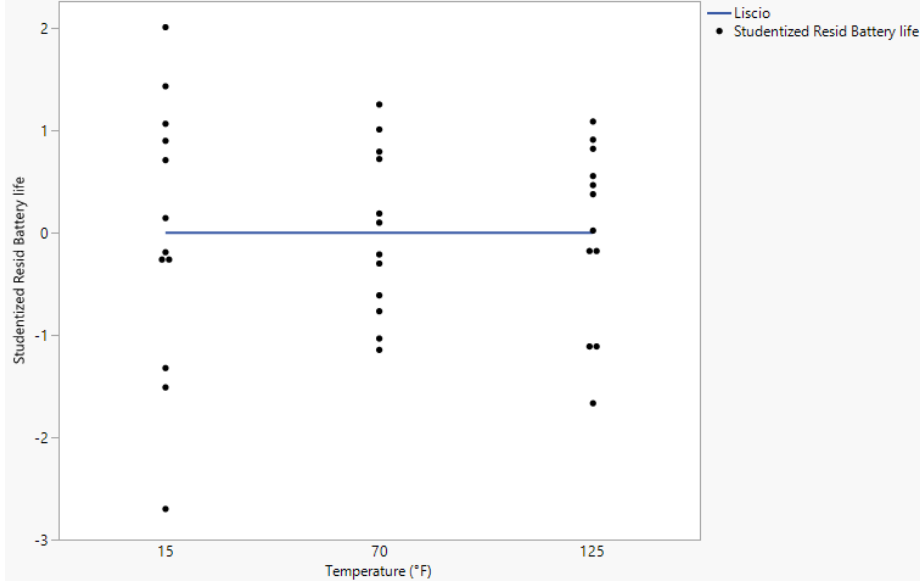
Studentized Resid Battery life vs. Predicted Battery life



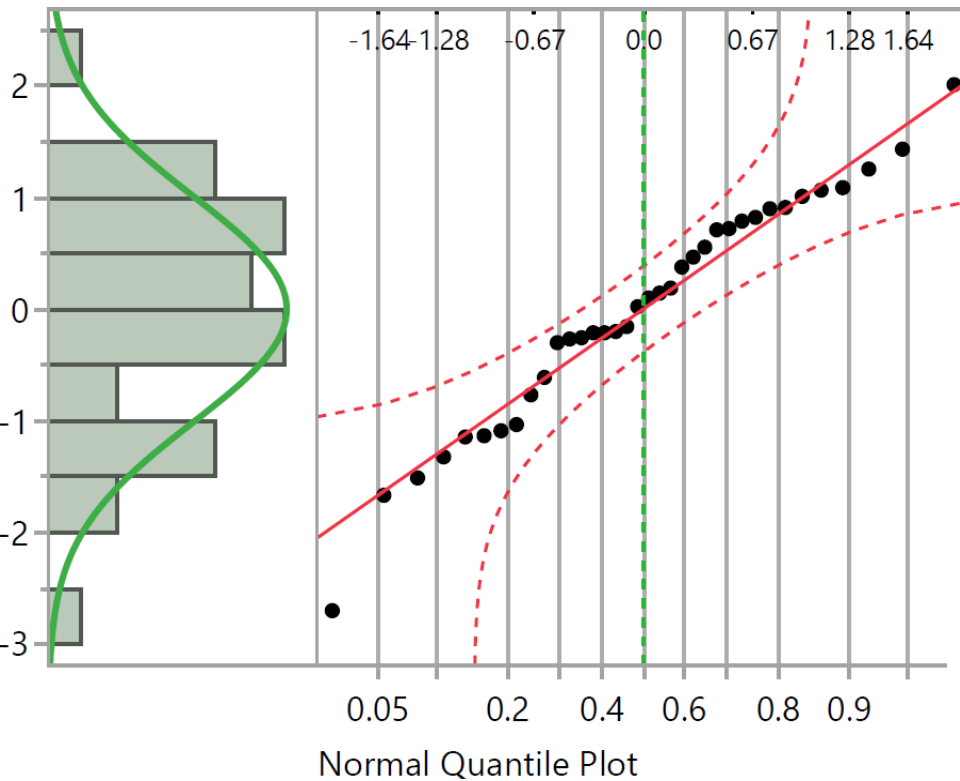
Studentized Resid Battery life vs. Material type



Studentized Resid Battery life rispetto a Temperature



Residual checking – Battery life



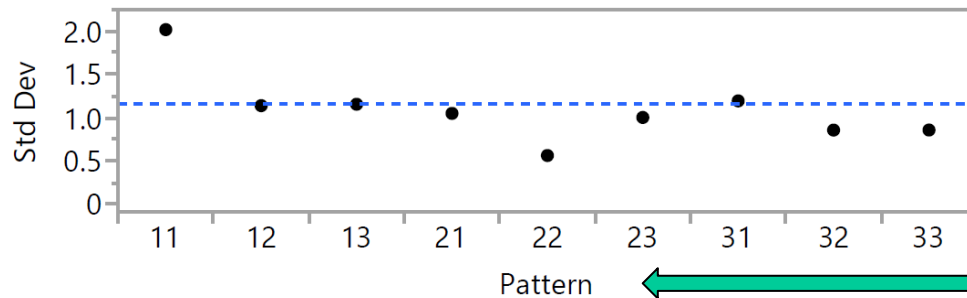
Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.976057	0.6117
	A ²	Simulated p-Value
Anderson-Darling	0.3403366	0.4864

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .

Test for Equal variances

Tests that the Variances are Equal



Fit Y by X

Put in X the **Pattern** column

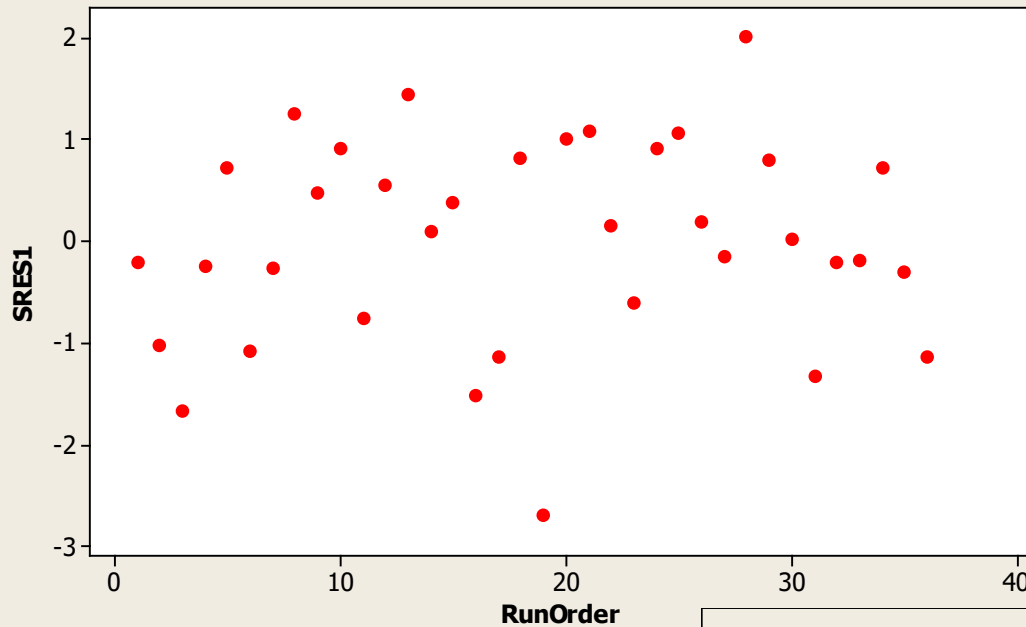
Or **Treatment**

Level	Count	Std Dev	MeanAbsDif	MeanAbsDif
			to Mean	to Median
11	4	2.015382	1.455326	1.455326
12	4	1.138371	0.788764	0.788764
13	4	1.154233	0.888749	0.888749
21	4	1.048683	0.899858	0.899858
22	4	0.562533	0.411046	0.411046
23	4	1.001815	0.722108	0.722108
31	4	1.193209	0.833202	0.822093
32	4	0.855925	0.644343	0.644343
33	4	0.856694	0.644343	0.644343

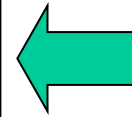
Test	F Ratio	DFNum	DFDen	Prob > F
O'Brien[.5]	0.9471	8	27	0.4958
Brown-Forsythe	0.7996	8	27	0.6081
Levene	0.9019	8	27	0.5289
Bartlett	0.6544	8	.	0.7321

Warning: Small sample sizes. Use Caution.

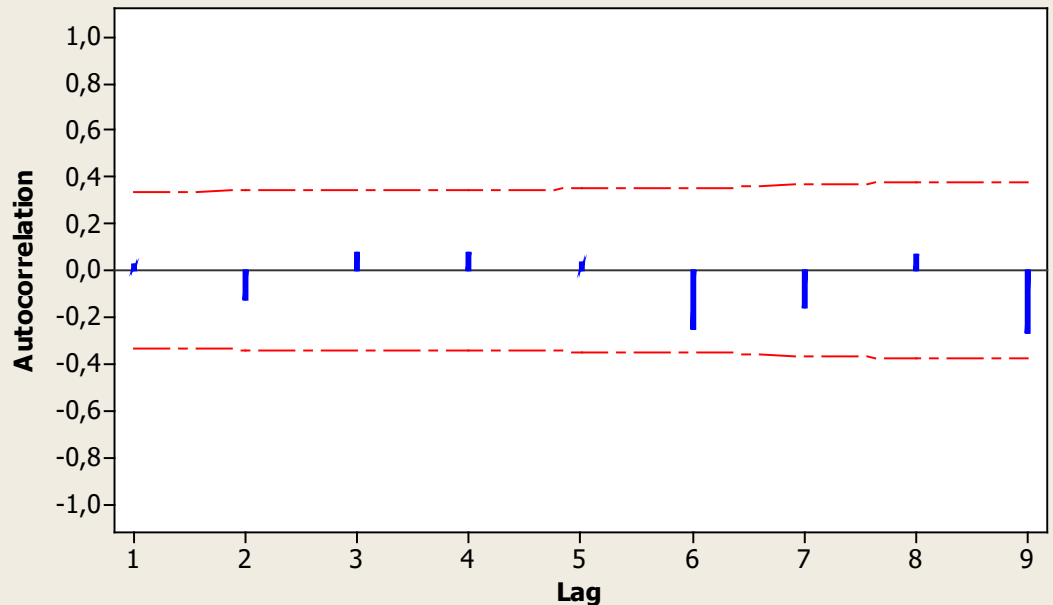
Scatterplot of SRES1 vs RunOrder



It's important to check the studentized-residual autocorrelation **if** you have the time order



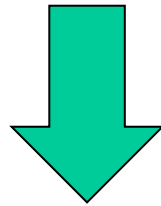
Autocorrelation Function for SRES1
(with 5% significance limits for the autocorrelations)



ANOVA for Factorial Designs

$$y_{ijk} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \hat{\tau\beta}_{ij} + e_{ijk}$$

$$e_{ijk} = y_{ijk} - \bar{y}_{ij}.$$



$$\sum_{i,j,k} (y_{ijk} - \hat{\mu})^2 = bn \sum_i \hat{\tau}_i^2 + an \sum_j \hat{\beta}_j^2 + n \sum_{i,j} \hat{\tau\beta}_{ij}^2 + \sum_{i,j,k} e_{ijk}^2$$

Note: Mixed products are null

$$\sum_{i,j,k} (y_{ijk} - \hat{\mu})^2 = \boxed{bn \sum_i \hat{\tau}_i^2} + \boxed{an \sum_j \hat{\beta}_j^2} + \boxed{n \sum_{i,j} \widehat{\tau\beta}_{ij}^2} + \boxed{\sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\cdot})^2}$$

$$SS_{TOT} = \boxed{SS_A} + \boxed{SS_B} + \boxed{SS_{AB}} + \boxed{SS_E}$$

$$abn - 1 = \boxed{(a - 1)} + \boxed{(b - 1)} + \boxed{(a - 1)(b - 1)} + \boxed{ab(n - 1)}$$

NB1 Orthogonal Design (Full Balanced Factorial Experiment)

NB2 SS_{TOT} Total Sum of Squares **Corrected**

We can prove that:

$$E(MS_A) = E\left(\frac{SS_A}{a-1}\right) = \sigma^2 + \Delta_A^2 = \sigma^2 + \frac{bn \sum_{i=1,a} \tau_i^2}{a-1}$$

$$E(MS_B) = E\left(\frac{SS_B}{b-1}\right) = \sigma^2 + \Delta_B^2 = \sigma^2 + \frac{an \sum_{j=1,b} \beta_j^2}{b-1}$$

$$E(MS_{AB}) = E\left(\frac{SS_{AB}}{(a-1)(b-1)}\right) = \sigma^2 + \Delta_{AB}^2 = \sigma^2 + \frac{n \sum_{i,j} \tau \beta_{ij}^2}{(a-1)(b-1)}$$

$$E(MS_E) = E\left(\frac{SS_E}{ab(n-1)}\right) = \sigma^2$$

ANOVA Table – Fixed Effects

Table 5-3 The Analysis of Variance Table for the Two-Factor Factorial, Fixed Effects Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
<i>A</i>	SS_A	$a - 1$	$MS_A = \frac{SS_A}{a - 1}$	$F_0 = \frac{MS_A}{MS_E}$
<i>B</i>	SS_B	$b - 1$	$MS_B = \frac{SS_B}{b - 1}$	$F_0 = \frac{MS_B}{MS_E}$
<i>AB</i>	SS_{AB}	$(a - 1)(b - 1)$	$MS_{AB} = \frac{SS_{AB}}{(a - 1)(b - 1)}$	$F_0 = \frac{MS_{AB}}{MS_E}$
Error	SS_E	$ab(n - 1)$	$MS_E = \frac{SS_E}{ab(n - 1)}$	
Total	SS_T	$abn - 1$		

$$SS_{TOT} = \sum_{i,j,k} y_{ijk}^2 - \frac{y_{...}^2}{abn}$$

$$SS_B = \frac{1}{an} \sum_j y_{.j.}^2 - \frac{y_{...}^2}{abn}$$

$$SS_A = \frac{1}{bn} \sum_i y_{i..}^2 - \frac{y_{...}^2}{abn}$$

$$SS_{AB} = \frac{1}{n} \sum_{i,j} y_{ij.}^2 - \frac{y_{...}^2}{abn} - SS_A - SS_B$$



$$SS_E = SS_{TOT} - SS_A - SS_B - SS_{AB}$$

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	8	59416.222	7427.03	10.9995
Error	27	18230.750	675.21	Prob > F
C. Total	35	77646.972		<.0001*

$$MSE = \hat{\sigma}^2 \equiv s^2 = 675.21$$

$$RMSE \equiv \hat{\sigma} \equiv s = \sqrt{MSE} = 25.98$$

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temperature	2	2	39118.722	28.9677	<.0001*
Material type	2	2	10683.722	7.9114	0.0020*
Temperature*Material type	4	4	9613.778	3.5595	0.0186*

Summary of Fit

RSquare	0.76521
RSquare Adj	0.695642
Root Mean Square Error	25.98486
Mean of Response	105.5278
Observations (or Sum Wgts)	36

$$5/3 = 1.67\%$$

$$\frac{SS_{MOD}}{SS_{TOT}} = 1 - \frac{SS_E}{SS_{TOT}}$$

$$1 - \frac{SS_E / df_E}{SS_{TOT} / df_{TOT}}$$

$$SS_{MOD} = SS_A + SS_B + SS_{AB} = SS_{TOT} - SS_E$$

Discuss how to compute the **p-value** and how to apply the **Bonferroni approximation** on the F Tests

Power of ANOVA F Test

Balanced Factorial Experiments

(Fixed Effects)

$$E\left(MS_{[]} \right) = \sigma^2 + \Delta_{[]}^2$$

$$\Delta_{[]}^2 = \frac{n * \text{Max} \left[\prod \left(\text{number of factor levels not in } [] \right), 1 \right]}{df_{[]}} \sum_{i,j,\dots} []_{ij..}^2$$

e.g. for **two factors** A (1:*a*), B (1:*b*) and ***n*** constant:

$$\Delta_B^2 = \frac{n}{(b-1)} \text{Max}[a, 1] \sum_j \beta_j^2 = \frac{na}{(b-1)} \sum_i \beta_j^2$$

$$\Delta_{AB}^2 = \frac{n}{(a-1)(b-1)} \text{Max}[0, 1] \sum_{i,j} \tau \beta_{ij}^2 = \frac{n}{(a-1)(b-1)} \sum_{i,j} \tau \beta_{ij}^2$$

$$\delta_{[]} = \frac{\Delta_{[]}^2}{\sigma^2} df_{[]}$$

$$\delta_B = \frac{na}{\sigma^2} \sum_{j=1,b} \beta_j^2$$

$$\delta_{AB} = \frac{n}{\sigma^2} \sum_{i,j} \tau \beta_{ij}^2$$

If **d** is the **minimum difference of interest**

$$\frac{d^2}{2} = \sum_{i,j..} []_{ij..}^2$$

$$\Delta_{[]}^2 = \frac{n * \text{Max} \left[\prod \left(\text{number of factor levels not in} [] \right), 1 \right]}{df_{[]}} \frac{d^2}{2}$$

Suppose **d ≥ 40** hours for **Temperature** (Factor B).

$$\frac{d}{\sigma} = 1.6$$

Suppose **σ²=25²** (Pilot experimentation)

$$\delta_{[]} = \frac{\Delta_{[]}^2}{\sigma^2} df_{[]} \qquad \delta_B = \frac{an}{2} \left(\frac{d}{\sigma} \right)^2 = \frac{3 * n}{2} \left(\frac{40}{25} \right)^2 = 3.84 * n$$

$$\alpha = 0.05 \qquad \text{Power} \geq 0.96$$

<i>n</i>	<i>df_B</i> = b-1	<i>df_E</i> = ab(n-1)	F _α	δ _B	Power
3	2	18	3.555	11.52	0.803
4	2	27	3.354	15.36	0.923
5	2	36	3.259	19.2	0.972



JMP computation

JMP does not make a **Main Effect** power computation in a multifactor context, but we can use a trick to approximate its value. For **factor B** you must split the data in ***b* groups** (in our case **3**) and consider for each group a number of replications equal to ***a***n*** (in our case **3*n***).

The result for ***n*=4** is practically equal **92.80** vs **92.30**

and the solution (**$P \geq 0.96$**) is equal

Power Explorer for ANOVA

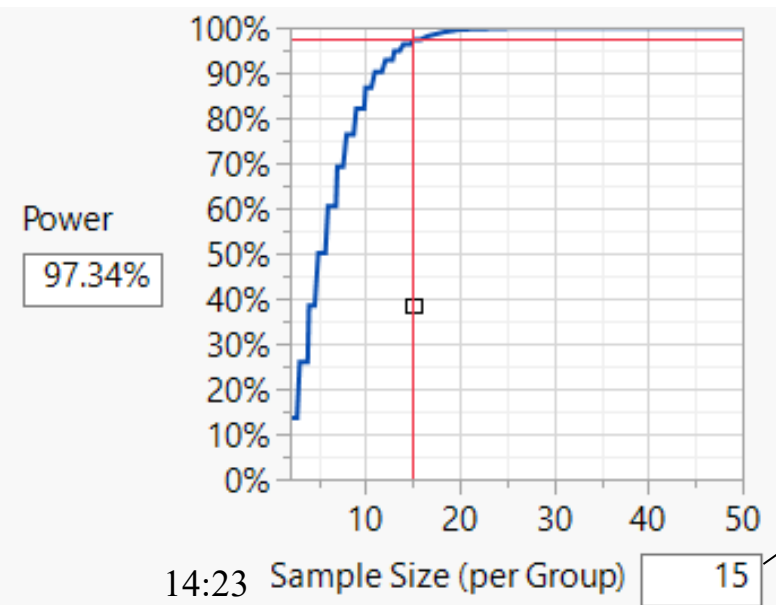
Explorer Settings

Test Specification

- ☐ Group means
- ☐ Between-group variance
- ☒ Maximum Difference
- ☒ Worst Case
- ☐ All but One

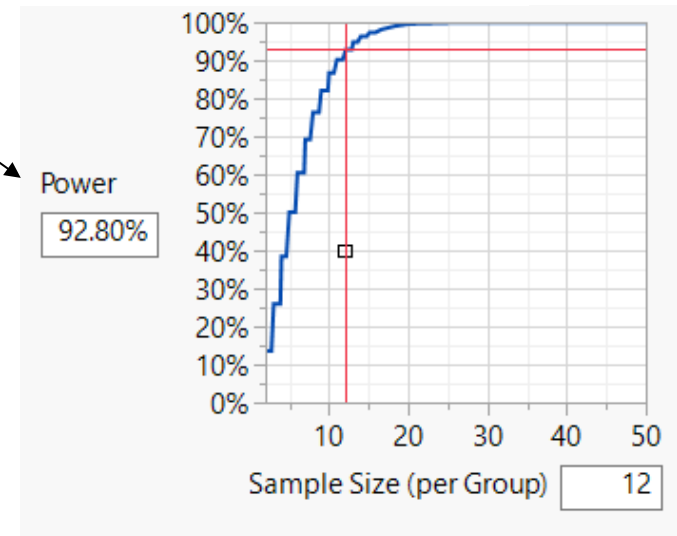
Preliminary Information

Alpha 0.05
Number of groups 3



$n=5$

QS-AA25-26



- And if **d ≥ 40** for **AB**?

$$\delta_{AB} = \frac{n}{2} \left(\frac{40}{25} \right)^2 = 1.28n$$

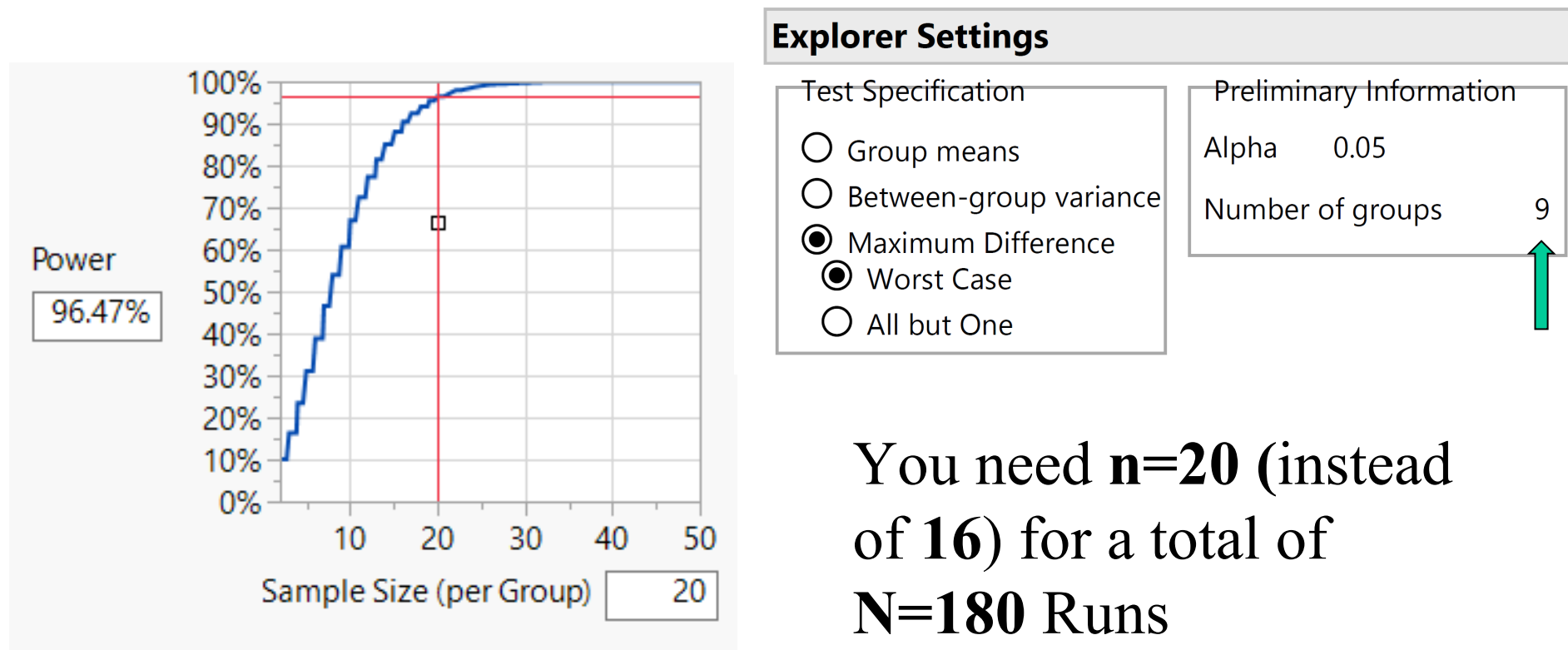
n	<i>df</i> _{AB} =(a-1)(b-1)	<i>df</i> _E =ab(n-1)	F _α	δ _{AB}	Power
4	4	27	2.7278	5.12	0.344
5	4	36	2.6335	6.40	0.443
...			...	...	...
14	4	117	2.4492	17.92	0.932
15	4	126	2.4436	19.20	0.949
 16	4	135	2.4387	20.48	0.962

We shifted from **N= 5*9=45** to **N=16*9=144** runs

JMP does not make this computation.

However JMP computes the power for the **Overall Model**.

In other words **JMP** considers all data cells as levels of a virtual single factor and makes a **One Factor ANOVA Power analysis**. This is the JMP output:



You need **n=20** (instead of **16**) for a total of **N=180** Runs

Reduced Model: ANOVA Table

(Battery Life example)

$$y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon$$

SS add *df* add but *MS* **do not** add (for Full Balanced Design)

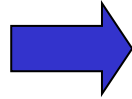
<i>Source</i>	<i>SS</i> _{Full}	<i>SS</i> _{RedMod}	<i>df</i>
A-material	10683.72	10683.72	2
B-temperature	39118.72	39118.72	2
AB	9613.78		4
Error	18230.75	27844.53	27
Total (corrected)	77646.97	77646.97	35

Red arrows indicate that the *SS* and *df* for the Error term are summed to 31, which is then added to the 4 degrees of freedom for the AB interaction to reach the total of 35 degrees of freedom.

One observation per cell

$$SS_E = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\cdot})^2 = \sum_{i,j} (y_{ij1} - y_{ij1})^2 = 0$$

$$df_E = ab(n-1) = 0$$



We can not estimate σ^2

But



$$E(MS_{AB}) = \sigma^2 + \frac{n \sum_{i,j} \tau \beta_{ij}^2}{(a-1)(b-1)}$$

IF we assume that the interaction effects are **zero**
we could use MS_{AB} as a σ^2 estimator

IF we can use a **known** or an **external estimate** of σ^2 if H_0 holds

$$df_{[]} \frac{MS_{[]}}{\sigma^2} \sim \chi^2(df_{[]})$$

New ANOVA table (Internal Estimate of MSE)

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
A	SS_A	$a-1$	MS_A	MS_A/MS_E
B	SS_B	$b-1$	MS_B	MS_B/MS_E
Residuals (they come from AB)	SS_{AB}	$(a-1)(b-1)$	$MS_{AB}=MS_E$ 	
Total (corrected)	SS_{TOT}	$ab-1$		

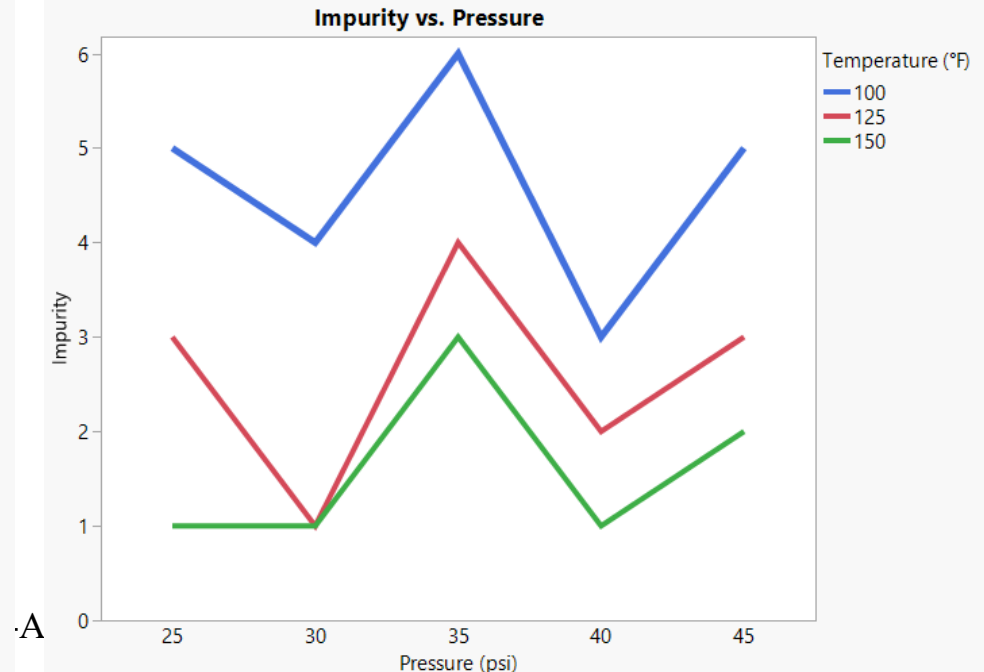
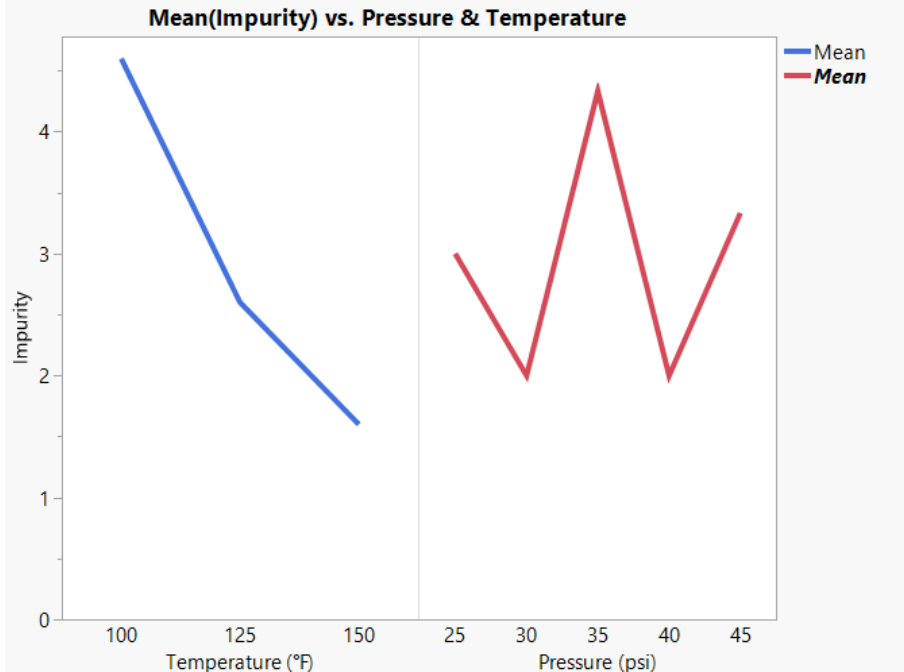
Remarks

- How plausible is the no-interaction hypothesis?
 - Interaction plot pattern
 - Tukey test for no-interaction (product form)
- What are the effects of a violation?
- The effect is to overestimate the true error variance and we have two cases:
 - If a model term is Significant -> ok
 - If a model term is Not-Significant -> doubt

Example 5.2 Impurity level

- The **impurity level** depends on two factors

	Pressure (psi)				
Temperature °F	25	30	35	40	45
100	5	4	6	3	5
125	3	1	4	2	3
150	1	1	3	1	2



ANOVA Table

Analysis of Variance

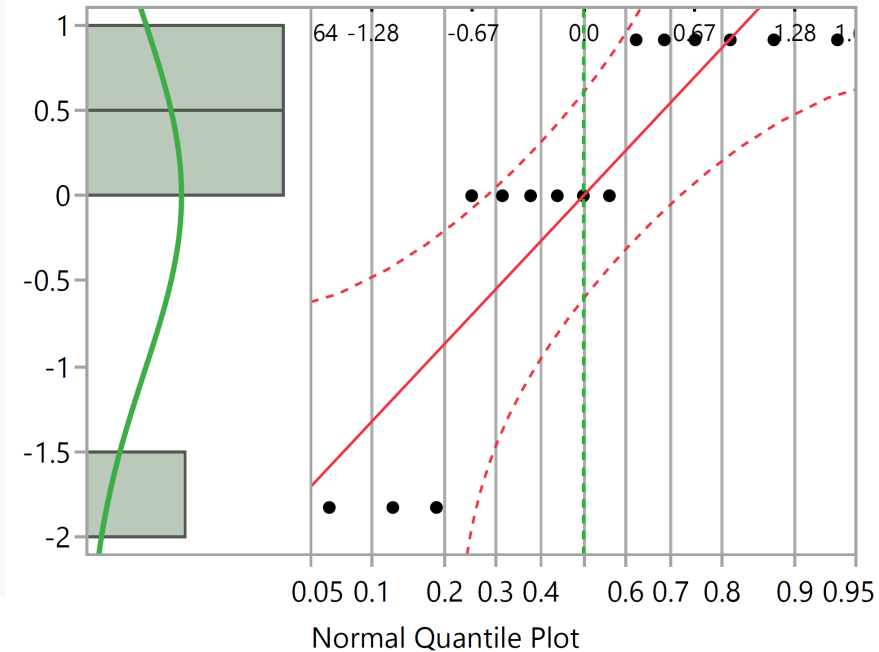
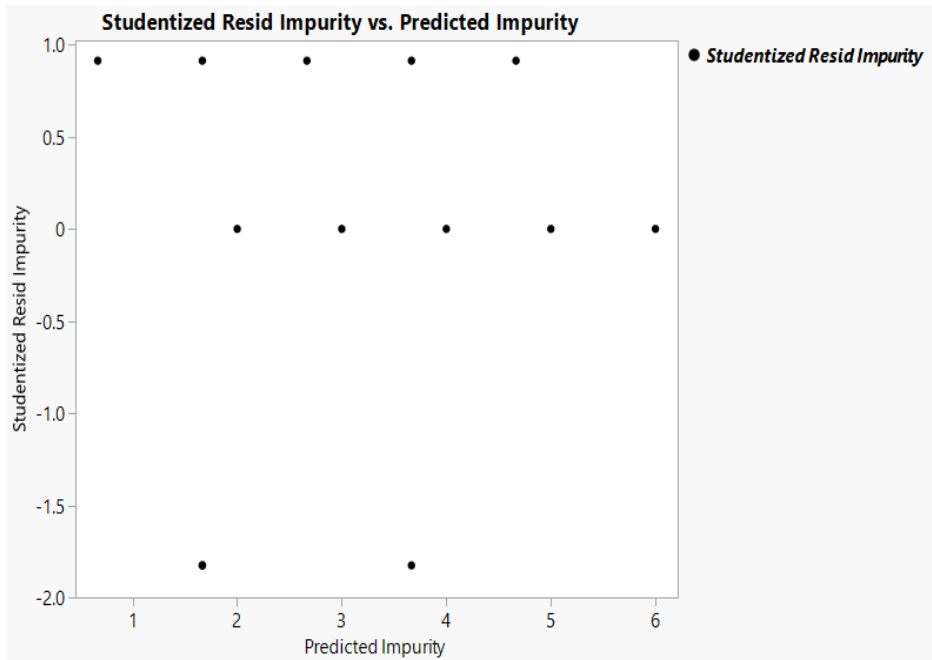
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	34.933333	5.82222	23.2889
Error	8	2.000000	0.25000	Prob > F
C. Total	14	36.933333		0.0001*

Summary of Fit

RSquare	0.945848
RSquare Adj	0.905235
Root Mean Square Error	0.5
Mean of Response	2.933333
Observations (or Sum Wgts)	15

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Pressure	4	4	11.600000	11.6000	0.0021*
Temperature	2	2	23.333333	46.6667	<.0001*



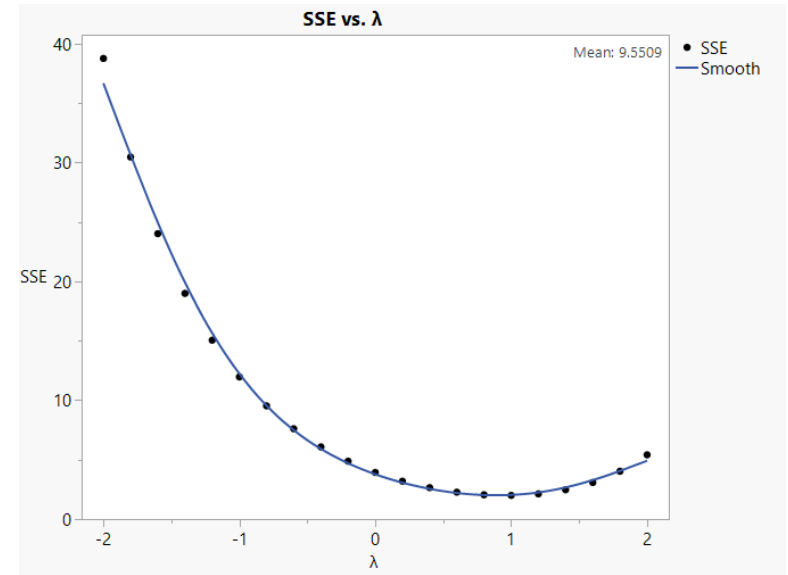
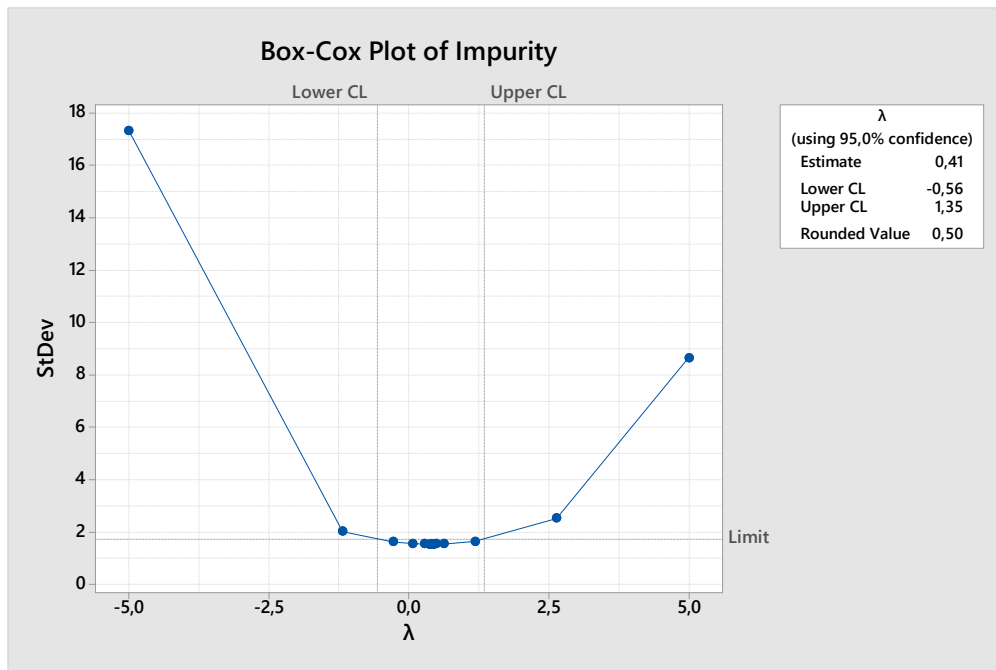
Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.7589397	0.0011*
	A ²	Simulated p-Value
Anderson-Darling	1.4898486	<.0001*

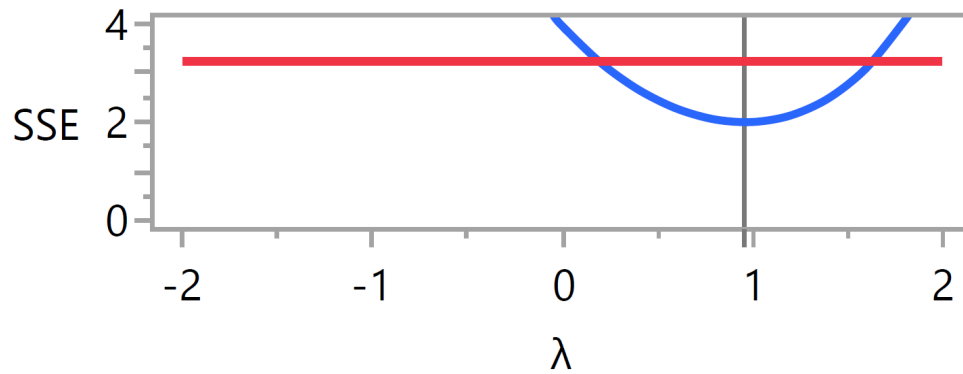
Note: Ho = The data is from the Normal distribution.

Small p-values reject Ho.

QS-AA25-26



Box-Cox Transformations



Best $\lambda = 0.958$

$$y^* = \sqrt{y}$$

Summary of Fit

RSquare	0.927284
RSquare Adj	0.872747
Root Mean Square Error	0.175768
Mean of Response	1.645217
Observations (or Sum Wgts)	15

Analysis of Variance

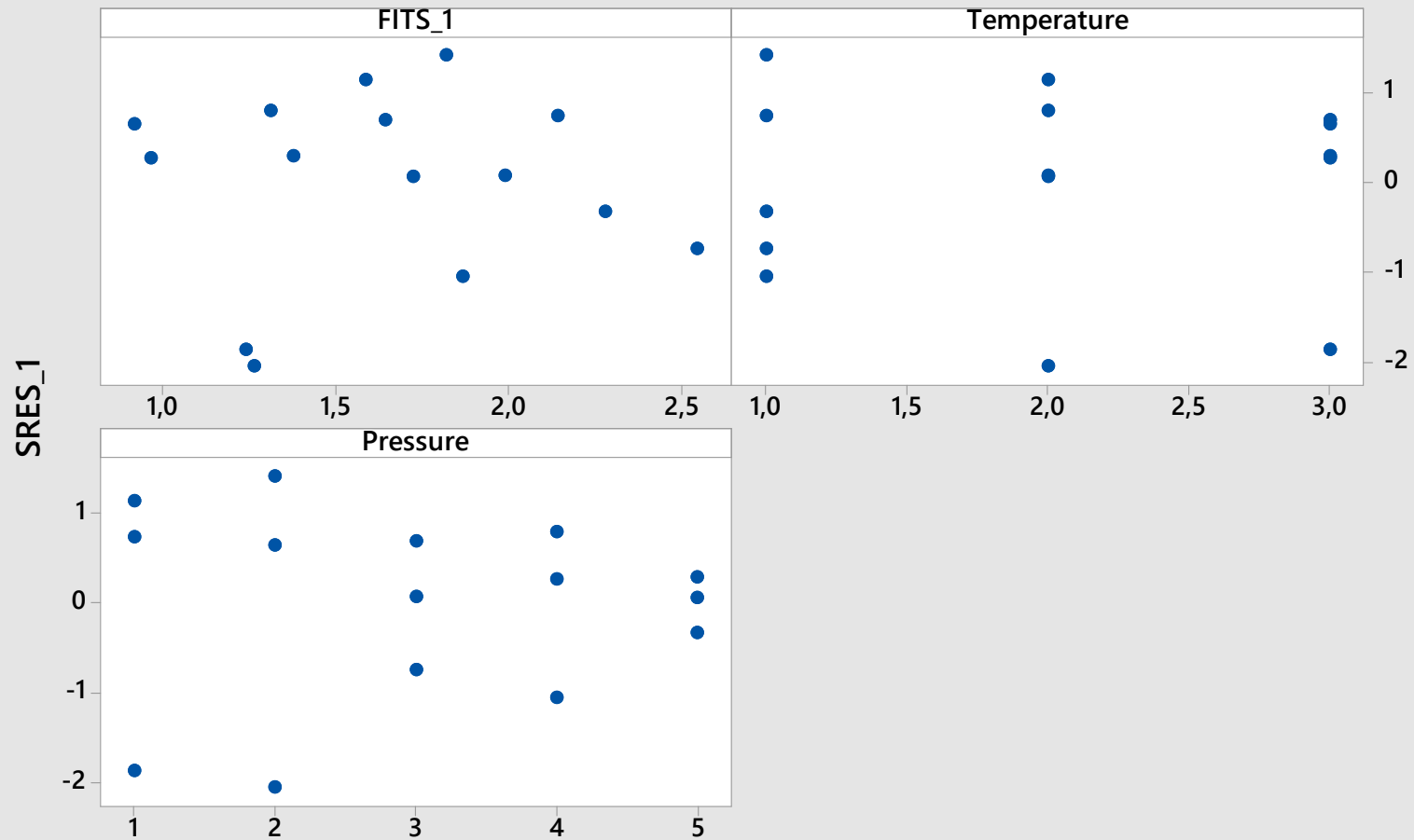
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	6	3.1517574	0.525293	17.0029
Error	8	0.2471545	0.030894	Prob > F
C. Total	14	3.3989119		0.0004*

Effect Tests

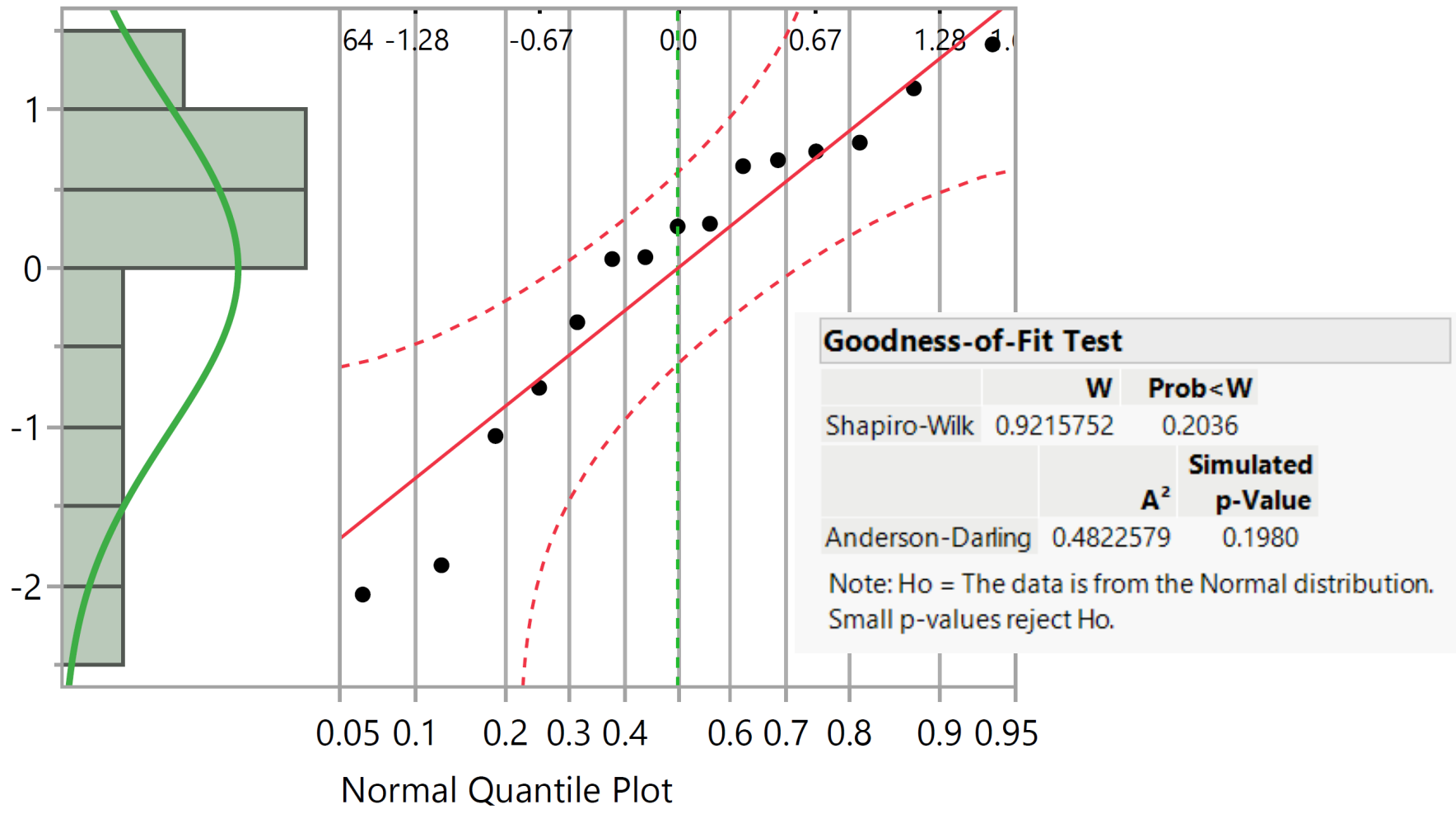
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Pressure	4	4	1.0837977	8.7702	0.0051*
Temperature	2	2	2.0679596	33.4683	0.0001*

Studentized Residual Plots

Scatterplot of SRES_1 vs FITS_1; Temperature; Pressure



Studentized Residuals vs. Studentized Residuals



Multiple mean comparisons (for $n=\text{const}$)

- Significant Interaction (Full Model)
- Additive Model (Reduced Model)

Significant Interaction (Full Model)

Each data cell represents a population: $y_{ijk} = \mu_{ij} + \varepsilon$
***ab* cells (means)**

Bonferroni $B_{\alpha_{FAM}} = t_{\alpha_{FAM}/2r}(df_E) \quad r = \frac{ab(ab-1)}{2}$

Tukey $T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(ab, df_E)$

Scheffé $S_{\alpha_{FAM}}^2 = (ab-1)F_{\alpha_{FAM}}(ab-1, df_E)$

$$\hat{V}(\hat{D}_{ij,i'j'}) = MS_E \left(\frac{1}{n_{ij}} + \frac{1}{n_{i'j'}} \right) \Rightarrow 2 \frac{MS_E}{n}$$

Additive model

$$\text{Model} \rightarrow y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon$$

- t-Bonferroni Method
- Tukey Method
- Scheffé Method

Factor A



$$D_{ii'} = \tau_i - \tau_{i'} \quad \hat{D}_{ii'} = \bar{y}_{i..} - \bar{y}_{i'..}$$

$$V(\hat{D}_{ii'}) = 2 \frac{\sigma^2}{bn} \quad \hat{V}(\hat{D}_{ii'}) = 2 \frac{MS_E}{bn}$$

TUKEY Coefficient $T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(a, df_E)$

$$\hat{D}_{ii'} - T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{ii'})} < D_{ii'} < \hat{D}_{ii'} + T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{ii'})}$$

$$\frac{|\hat{D}_{ii'}|}{\sqrt{\hat{V}(\hat{D}_{ii'})}} > T_{\alpha_{FAM}} \text{ reject } H_0$$

Factor B

$$D_{jj'} = \beta_j - \beta_{j'} \quad \hat{D}_{jj'} = \bar{y}_{\cdot j} - \bar{y}_{\cdot j'}$$

$$V(\hat{D}_{jj'}) = 2 \frac{\sigma^2}{an} \quad \hat{V}(\hat{D}_{jj'}) = 2 \frac{MS_E}{an}$$

TUKEY coefficient $T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(b, df_E)$

$$\hat{D}_{jj'} - T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{jj'})} < D_{jj'} < \hat{D}_{jj'} + T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{jj'})}$$

$$\frac{|\hat{D}_{jj'}|}{\sqrt{\hat{V}(\hat{D}_{jj'})}} > T_{\alpha_{FAM}} \text{ reject } H_0$$

Scheffé Method

$$L = \sum_i c_i \mu_i \quad \text{Factor A;}$$

$$\hat{L} = \sum_i c_i \bar{y}_{i..}$$

$$\hat{V}(\hat{L}) = \frac{MS_E}{bn} \sum_i c_i^2$$

$$L = \sum_j c_j \mu_j \quad \text{Factor B}$$

$$\hat{L} = \sum_j c_j \bar{y}_{.j.}$$

$$\hat{V}(\hat{L}) = \frac{MS_E}{an} \sum_j c_j^2$$

$$\hat{L} - S_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{L})} \leq L \leq \hat{L} + S_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{L})}$$

$$S_{\alpha_{FAM}}^2 = (b-1)F_{\alpha_{FAM}}(b-1, df_E)$$

$$S_{\alpha_{FAM}}^2 = (a-1)F_{\alpha_{FAM}}(a-1, df_E)$$

If we are interested in a simultaneous confidence interval for A **and** B, we can use:

a) **t-Bonferroni** method on all pair mean comparisons

b) **Bonferroni** method to join the **two sets** of intervals obtained by **Tukey** or **Scheffé** methods

$$\alpha = \frac{\alpha_{FAM}}{2}$$

c) **Scheffé** Method by putting

$$S_{\alpha_{FAM}}^2 = (a + b - 2)F_{\alpha_{FAM}}(a + b - 2, df_E)$$

General Full Balanced Factorial Design (K Factors)

- Always randomize the runs
- The math is the same
- ANOVA splits the Sum of Squares as usual

(**Note: this is true only for Full Balanced Model**)

$$SS_T = SS_A + SS_B + \cdots + SS_{AB} + SS_{AC} + \cdots \\ + SS_{ABC} + \cdots + SS_{AB \cdots K} + SS_E$$

- # of Model terms (excluding the mean) $\rightarrow 2^K - 1$
- Let's consider a **3 Factors Factorial Design**

Parameter Estimate

- Statistical Model (**fixed effects**)

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + \tau\beta_{ij} + \tau\gamma_{ik} + \beta\gamma_{jk} + \tau\beta\gamma_{ijk} + \varepsilon$$
$$i = 1, a \quad j = 1, b \quad k = 1, c \quad l = 1, n$$

- OLS estimates (constant n and additional constraints)

$$\hat{\mu} = \frac{y_{\dots}}{abcn} \quad \hat{\tau}_i = \bar{y}_{i\dots} - \hat{\mu} \quad \hat{\beta}_j = \bar{y}_{\cdot j \cdot} - \hat{\mu} \quad \hat{\gamma}_k = \bar{y}_{\cdot \cdot k} - \hat{\mu}$$

$$\hat{\tau\beta}_{ij} = \bar{y}_{ij \cdot} - \left(\hat{\mu} + \hat{\tau}_i + \hat{\beta}_j \right)$$

$$\hat{\tau\beta\gamma}_{ijk} = \bar{y}_{ijk \cdot} - \left[\hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \hat{\gamma}_k + \hat{\tau\beta}_{ij} + \hat{\tau\gamma}_{ik} + \hat{\beta\gamma}_{jk} \right]$$

Sum of Squares decomposition

- If the Design is orthogonal (Full Balanced Factorial Experiment)

$$y_{ijkl} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \hat{\gamma}_k + \widehat{\tau\beta}_{ij} + \widehat{\tau\gamma}_{ik} + \widehat{\beta\gamma}_{jk} + \widehat{\tau\beta\gamma}_{ijk} + e_{ijkl}$$

- Using the same procedure we saw:

$$\sum_{i,j,k,l} (y_{ijkl} - \hat{\mu})^2 = \sum_{i,j,k,l} \hat{\tau}_i^2 + \sum_{i,j,k,l} \hat{\beta}_k^2 + \dots + \sum_{i,j,k,l} \widehat{\tau\beta\gamma}_{ijk}^2 + \sum_{i,j,k,l} e_{ijkl}^2$$

$$\sum_{i,j,k,l} (y_{ijkl} - \hat{\mu})^2 = bcn \sum_i \hat{\tau}_i^2 + acn \sum_j \hat{\beta}_k^2 + \dots + n \sum_{i,j,k} \widehat{\tau\beta\gamma}_{ijk}^2 + \sum_{i,j,k,l} e_{ijkl}^2$$

- The SSs are: **Please Check the formulas**

$$SS_{TOT} = \sum_{i,j,k,l} \left(y_{ijkl} - \hat{\mu} \right)^2 = \sum_{i,j,k,l} y_{ijkl}^2 - \frac{y_{....}^2}{abcn}$$

$$SS_A = bcn \sum_i \hat{\tau}_i^2 = \sum_i \frac{y_{i...}^2}{bcn} - \frac{y_{....}^2}{abcn} \quad \text{etc.}$$

$$SS_{AB} = cn \sum_{i,j} \hat{\tau\beta}_{ij}^2 = \sum_{i,j} \frac{y_{ij..}^2}{cn} - \frac{y_{....}^2}{abcn} \quad \text{etc.}$$

$$SS_{ABC} = n \sum_{i,j,k} \hat{\tau\beta\gamma}_{ijk}^2 = \sum_{i,j,k} \frac{y_{ijk.}^2}{n} - \frac{y_{....}^2}{abcn}$$

$$SS_E = \sum_{i,j,k,l} e_{ijkl}^2 = \sum_{i,j,k,l} \left(y_{ijkl} - \bar{y}_{ijk.} \right)^2$$

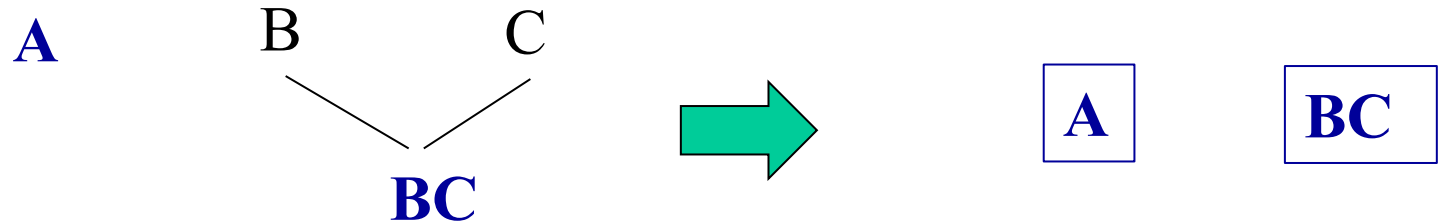
Three factors table

$$y_{ijkl} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \hat{\gamma}_k + \hat{\tau\beta}_{ij} + \hat{\tau\gamma}_{ik} + \hat{\beta\gamma}_{jk} + \hat{\tau\beta\gamma}_{ijk} + e_{ijkl}$$

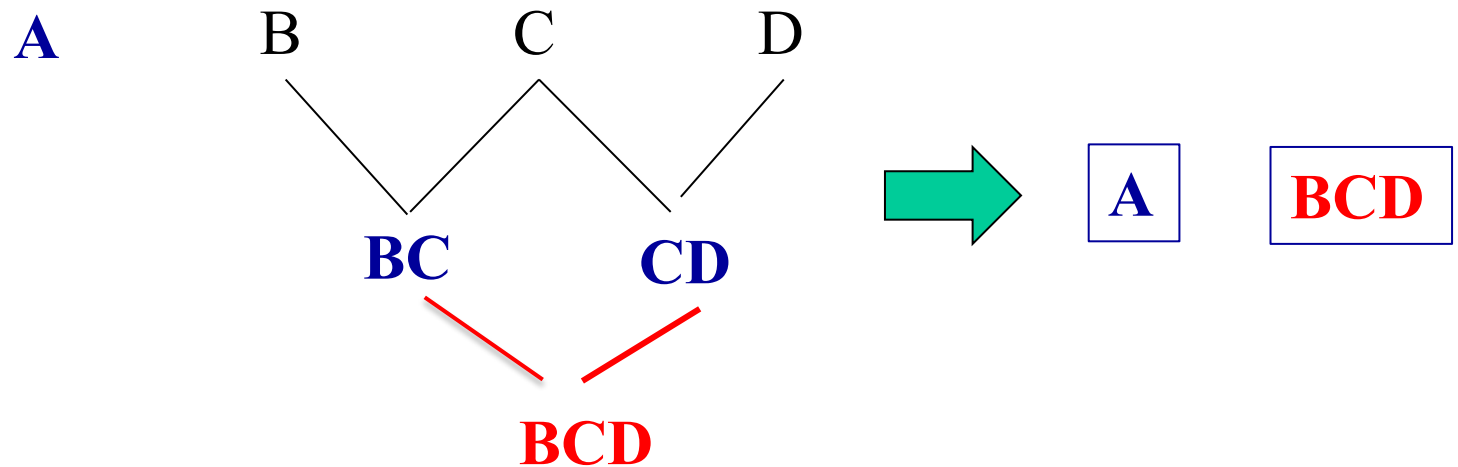
<i>A</i>	SS_A	$(a-1)$	MS_A	$\sigma^2 + \frac{bcn}{(a-1)} \sum_i \tau_i^2$
<i>B</i>	SS_B	$(b-1)$	MS_B	$\sigma^2 + \frac{acn}{(b-1)} \sum_j \beta_j^2$
<i>C</i>	SS_C	$(c-1)$	MS_C	$\sigma^2 + \frac{abn}{(c-1)} \sum_k \gamma_k^2$
<i>AB</i>	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	$\sigma^2 + \frac{cn}{(a-1)(b-1)} \sum_{i,j} \tau\beta_{ij}^2$
<i>BC</i>	SS_{BC}	$(b-1)(c-1)$	MS_{BC}	$\sigma^2 + \frac{an}{(b-1)(c-1)} \sum_{j,k} \beta\gamma_{jk}^2$
<i>AC</i>	SS_{AC}	$(a-1)(c-1)$	MS_{AC}	$\sigma^2 + \frac{bn}{(a-1)(c-1)} \sum_{i,k} \tau\gamma_{ik}^2$
<i>ABC</i>	SS_{ABC}	$(a-1)(b-1)(c-1)$	MS_{ABC}	$\sigma^2 + \frac{n}{(a-1)(b-1)(c-1)} \sum_{i,j,k} \tau\beta\gamma_{ijk}^2$
<i>Error</i>	SS_E	$abc(n-1)$	MS_E	σ^2
<i>Total</i>	SS_{TOT}	$abcn-1$		

Multiple comparisons

- **Example 1**



- **Example 2**



Power of ANOVA F Test Balanced Factorial Experiments (Fixed Effects)

$$E\left(MS_{[]} \right) = \sigma^2 + \Delta_{[]}^2$$

$$\delta_{[]} = \frac{\Delta_{[]}^2}{\sigma^2} df_{[]}$$

$$\Delta_{[]}^2 = \frac{n * \text{Max} \left[\prod \left(\text{number of factor levels not in } [] \right), 1 \right]}{df_{[]}} \sum_{i,j,\dots} []_{ij..}^2$$

- e.g. let us consider **4 factors A** (1:a), **B** (1:b), **C** (1:c), **F** (1:f)
- For the **ABC** term and constant **n** :

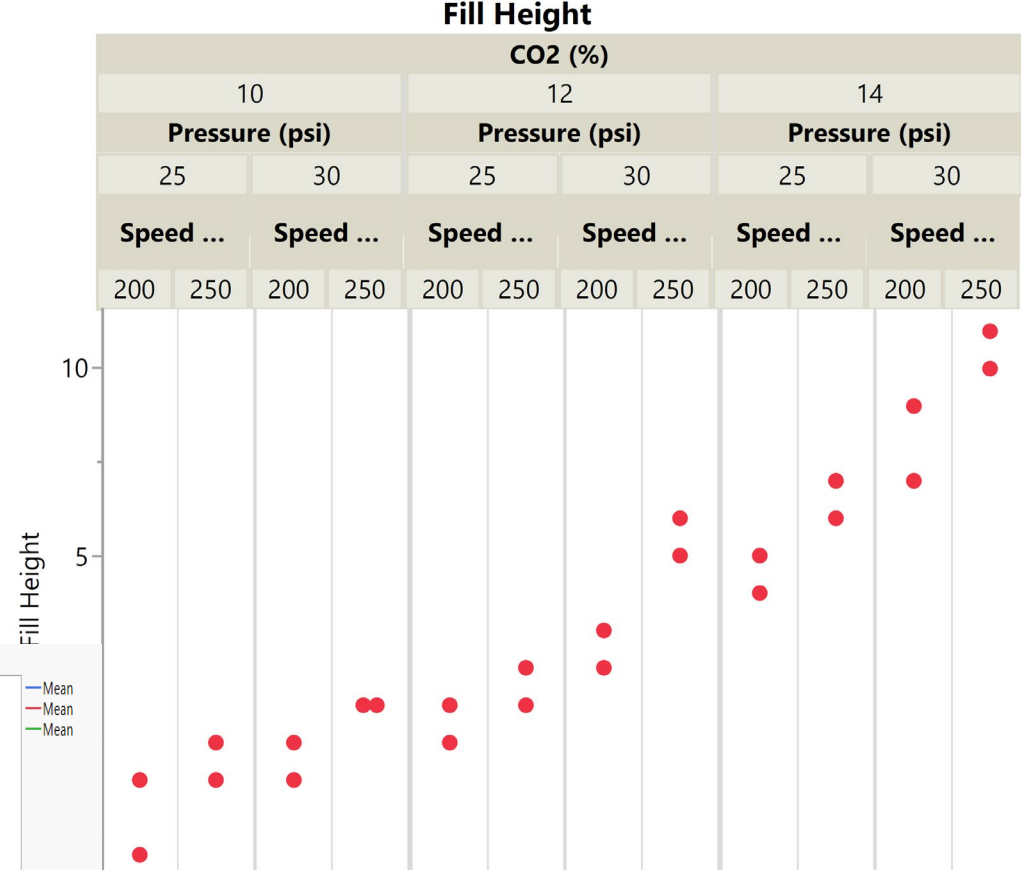
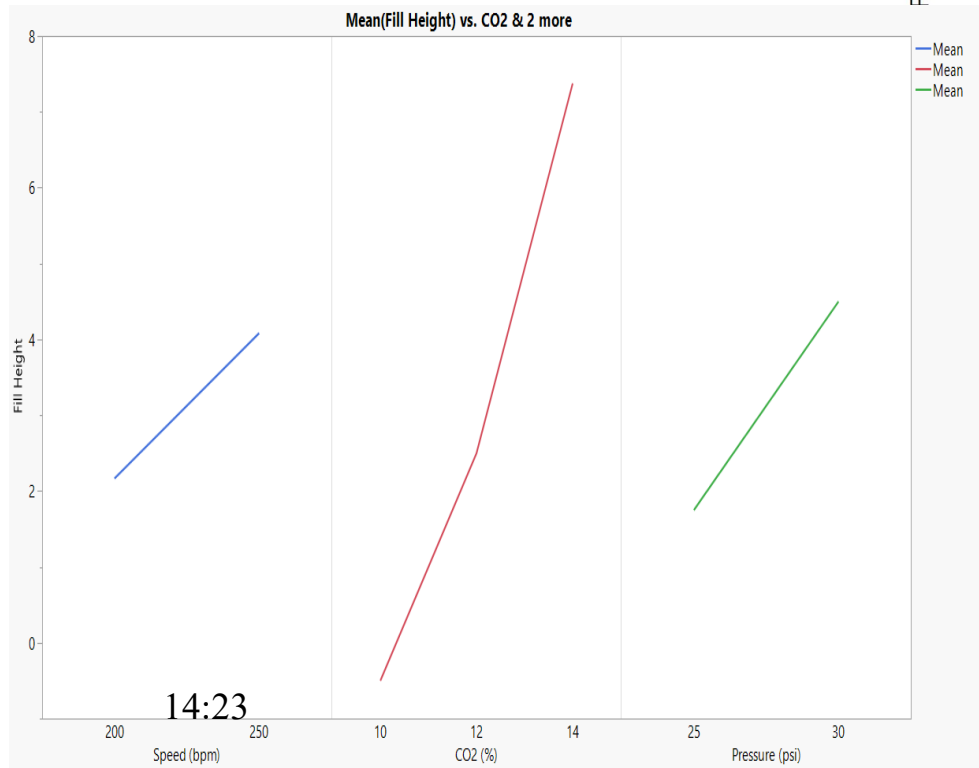
$$\Delta_{ABC}^2 = \frac{nf}{(a-1)(b-1)(c-1)} \sum_{\substack{i=1,a \\ j=1,b \\ k=1,c}} \alpha\beta\gamma_{ijk}^2 \quad \delta_{ABC} = \frac{nf}{\sigma^2} \sum_{\substack{i=1,a \\ j=1,b \\ k=1,c}} \alpha\beta\gamma_{ijk}^2 = \left(\frac{d}{\sigma} \right)^2 \frac{nf}{2}$$

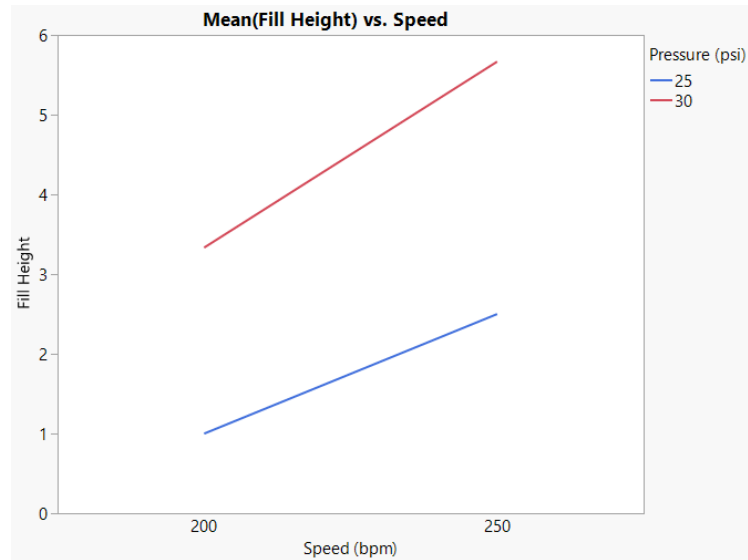
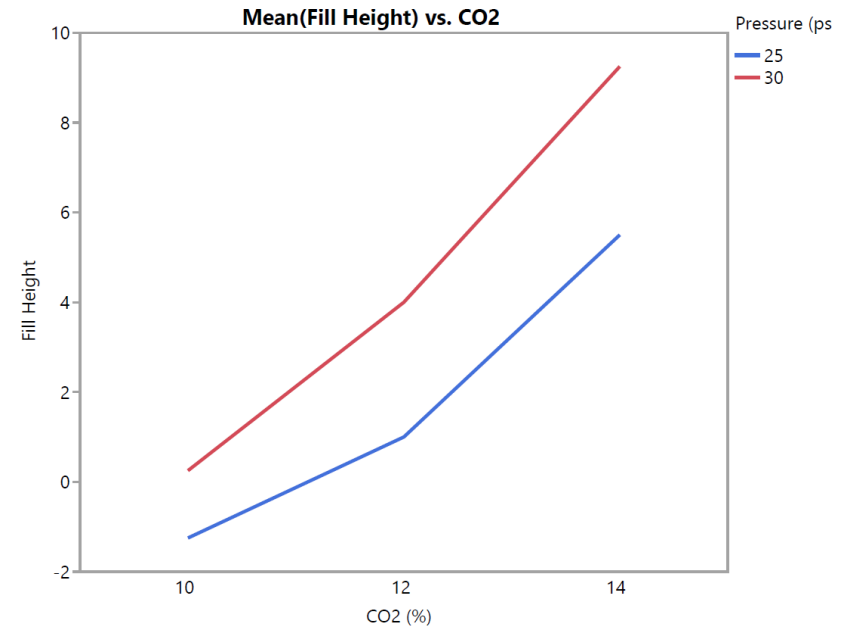
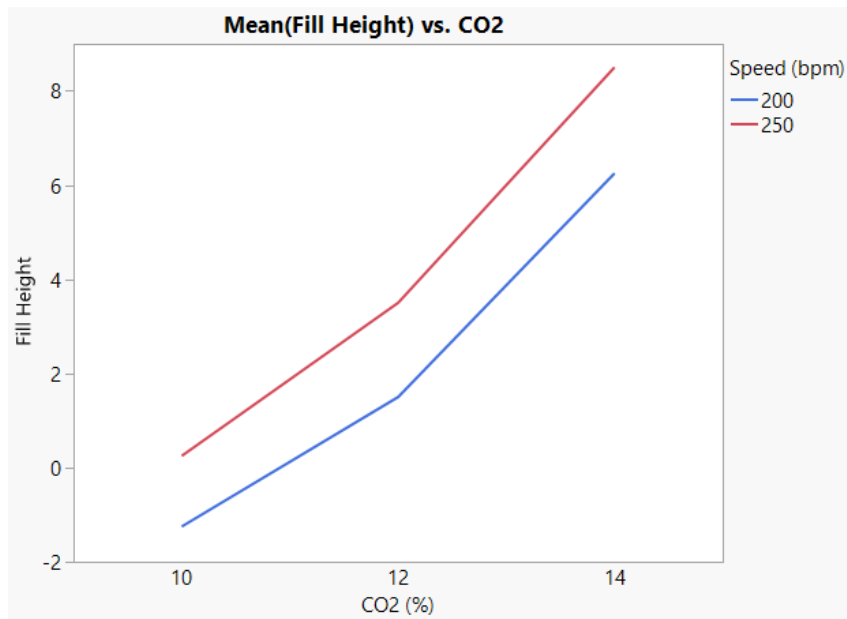
The Soft Drink Bottling Problem

Example 5.3

- A soft drink bottler is interested in obtaining a more uniform **fill heights** compared to the target value.
- He/she suspects that three operating parameters influence the Height **CO₂ %, operating pressure** in the filler and **line speed**
 - **CO₂** (%) 3 levels: 10 12 14
 - **pressure** (psi) 2 levels : 25 30
 - **Line speed** (bpm) 2 levels : 200 250
 - **Replications** 2

Data-set is in the book Exploratory plots





Interaction plots

Summary of Fit

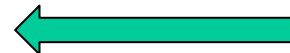
RSquare	0.974749
RSquare Adj	0.951603
Root Mean Square Error	0.841625
Mean of Response	3.125
Observations (or Sum Wgts)	24

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	11	328.12500	29.8295	42.1123
Error	12	8.50000	0.7083	Prob > F
C. Total	23	336.62500		<.0001*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
CO2	2	2	252.75000	178.4118	<.0001*
Pressure	1	1	45.37500	64.0588	<.0001*
Speed	1	1	22.04167	31.1176	0.0001*
CO2*Pressure	2	2	5.25000	3.7059	0.0558
CO2*Speed	2	2	0.58333	0.4118	0.6715
Pressure*Speed	1	1	1.04167	1.4706	0.2486
CO2*Pressure*Speed	2	2	1.08333	0.7647	0.4869

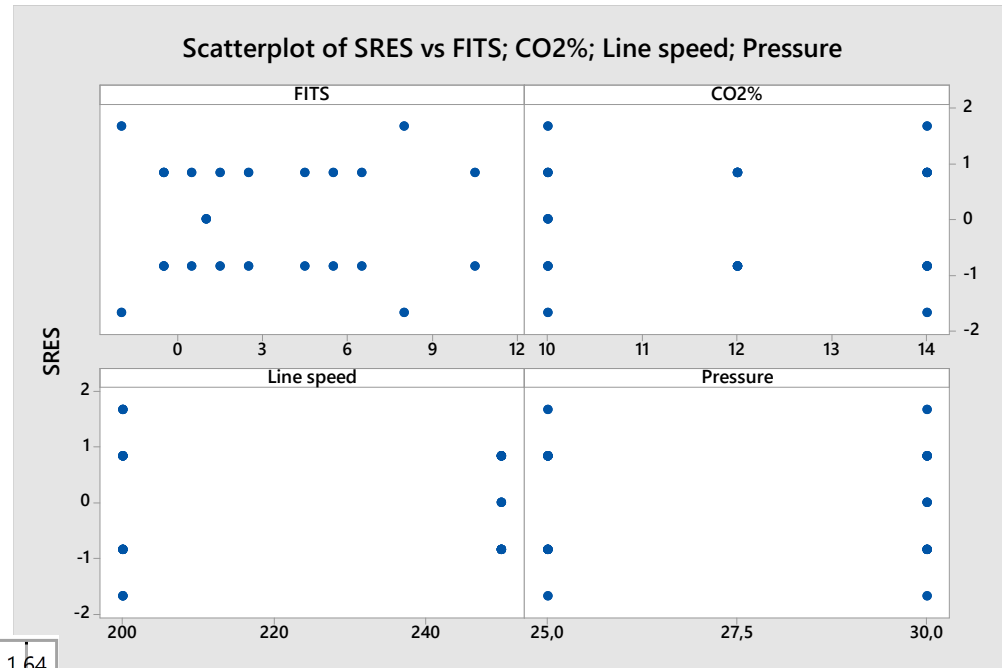
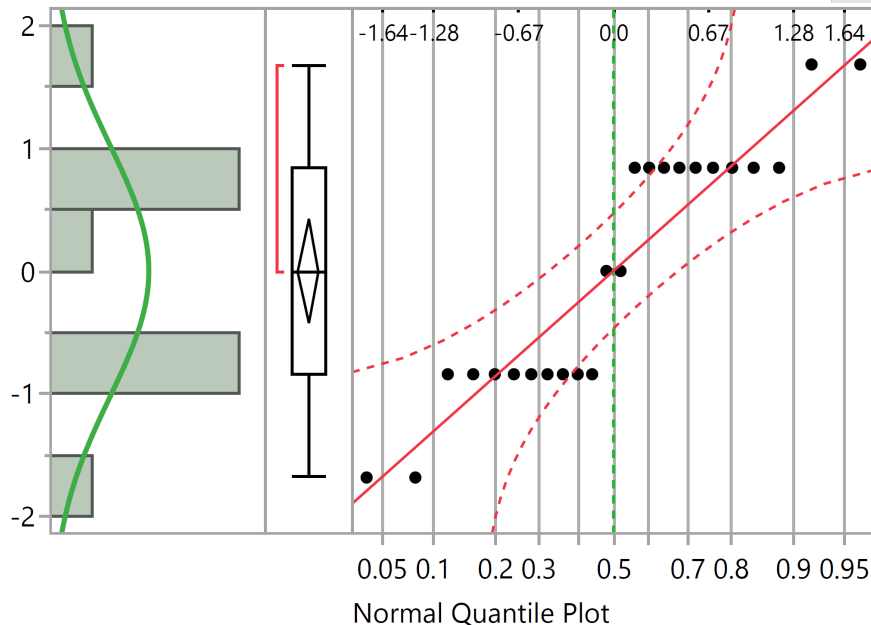


SResidual checking (residuals from the Full Model)

Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.8632162	0.0039*
	A ²	Simulated p-Value
Anderson-Darling	1.631933	<.0001*

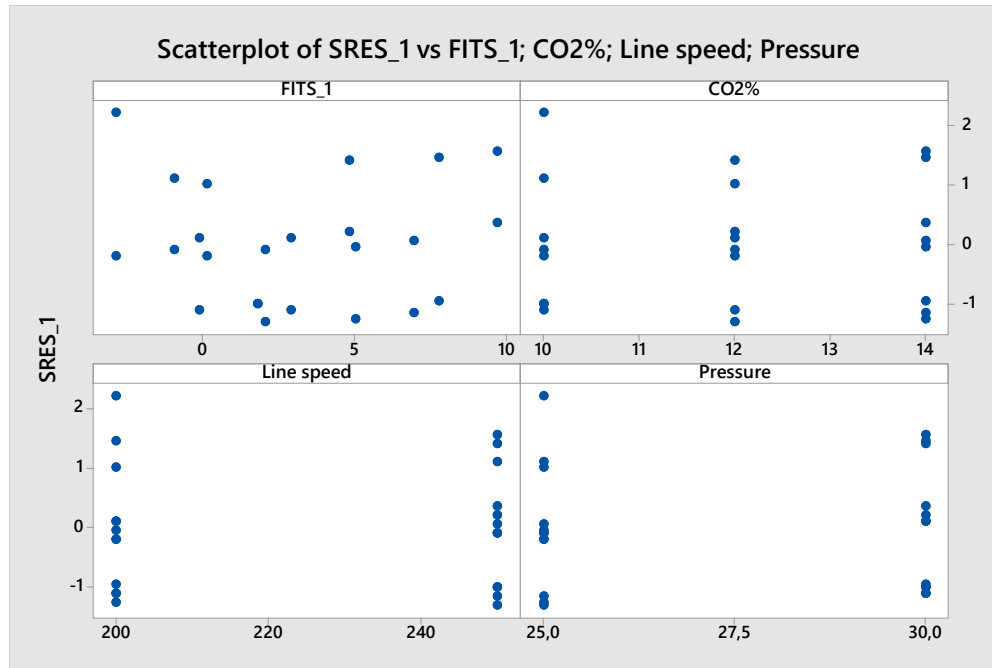
Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.



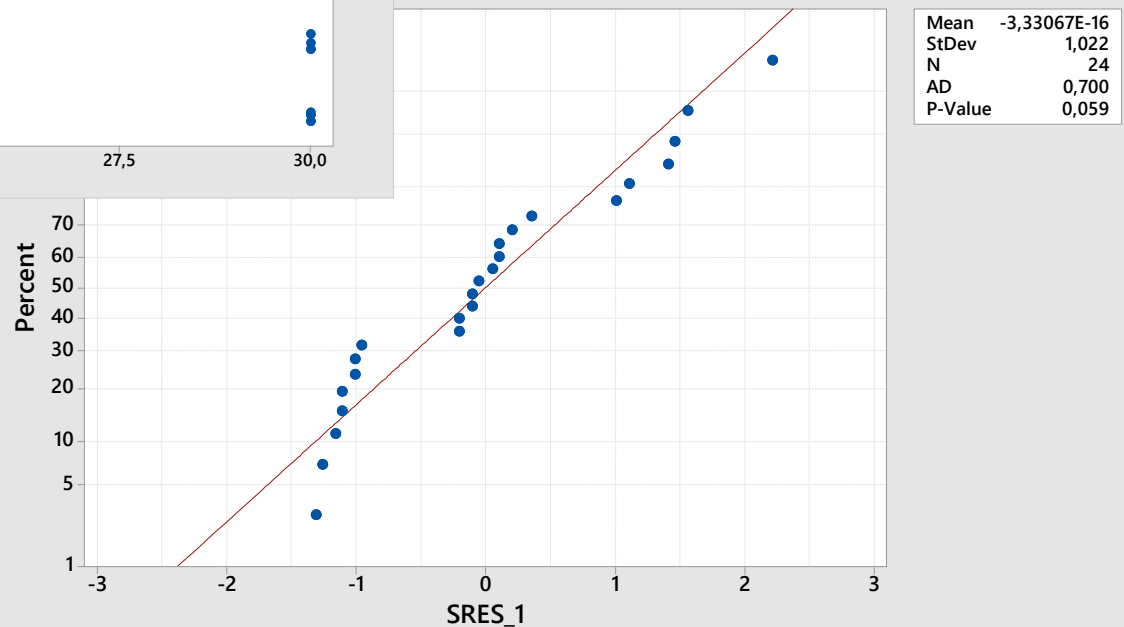
Should we worry?

This is called
Overfitting

Reduced Model (additive model)



Probability Plot of SRES_1
Normal



Power for Main Effect A

Let us compute the power for **Main effect A** (CO2 perc)

The exact computation gives for $d/\sigma=1$ and $n=2$, $P=0.33$

Explorer Settings

Test Specification

- ☐ Group means
- ☐ Between-group variance
- ☒ Maximum Difference

☒ Worst Case

☐ All but One

Preliminary Information

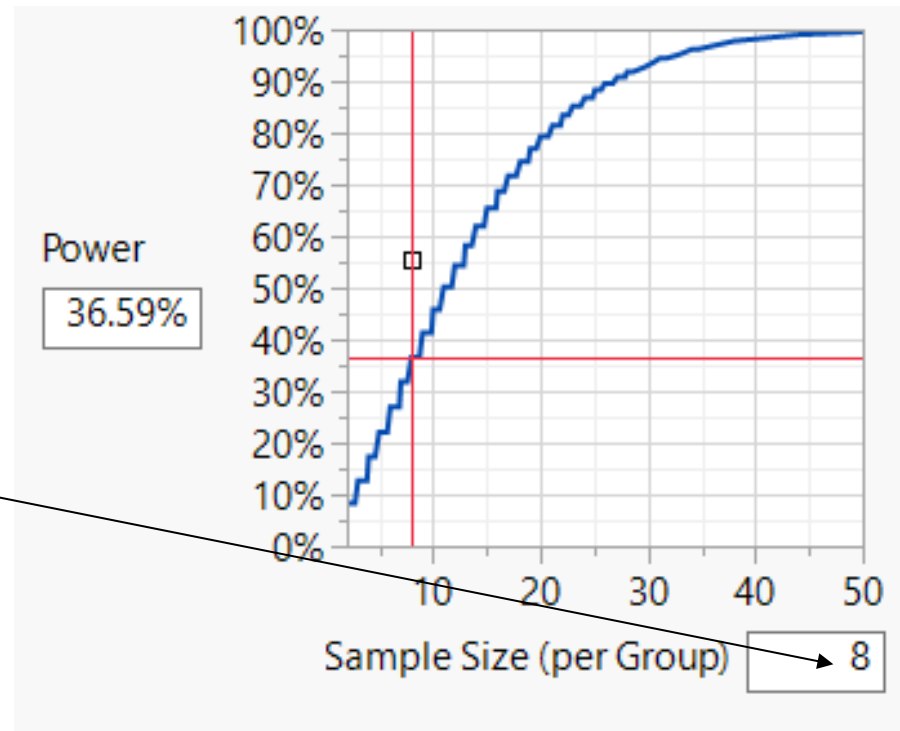
Alpha

Number of groups

3 levels of CO2

$$b*c*n=2*2*2=8$$

JMP gives $P=0.37$



Power for AB interaction

Let us compute the power for the **AB interaction** using $n=2$ replicates.

The exact computation gives **0.18**

If we use JMP with the trick.

The JMP computation gives **0.127**

Explorer Settings

Test Specification

- ☐ Group means
- ☐ Between-group variance
- ☒ Maximum Difference
 - ☒ Worst Case
 - ☐ All but One

Preliminary Information

Alpha 0.05
Number of groups 6

$ab=6$

$cn=4$

Power
12.74%

